Mary Anne Panoyan, M.Sc.

Ph.D. Candidate, Collaborative Specialization in Global Health
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Professional Summary

My research uses large-scale genomic data and genetic epidemiological methods to investigate the genetic and modifiable risk factors underlying brain health and chronic disease. My interests include complex-trait genomics, population genetics, and causal inference. I also have a passion for equitable pedagogy and incorporate Universal Design for Learning (UDL) principles into my teaching for a student–centred approach.

Summary of Skills

Teaching: University-Level Instruction, Experiential Learning, Evidence-Based Pedagogy, Student-Centred Learning, Course Design and Development, Quercus (Canvas)
Research: Genetic Epidemiology, Genome-Wide Association Studies (GWAS), Mendelian Randomization, Population Genetics, Tandem Repeat Analysis, Genomic Structural Equation Modelling, Linkage Disequilibrium Score Regression, Survival Analysis, Next-Generation Sequencing, Clinical Research, UK Biobank, All of Us
Programming & Data Analysis: Python, R, Bash, Linux, Git, PLINK, LDSC, GenomicSEM, TwoSampleMR, TensorFlow, Keras, Machine Learning (Regression, Random Forests, Neural Networks), Principal Component Analysis

Education

Doctoral Degree (Ph.D.) Biological Anthropology with a Collaborative Specialization in Global Health — 2021 - Expected August 2027
Department of Anthropology | Krembil Centre for Neuroinformatics | University of Toronto | Toronto, ON

- **Cumulative GPA**: 3.9/4.0
- **Supervisors**: Esteban Parra & Daniel Felsky
- **Doctoral Dissertation's Title**: Genomic and Causal Approaches to Understanding Genetic Determinants of Brain Health and Alzheimer's Disease
- **Relevant Coursework**: ANT3047 Evolutionary Anthropology | ANT3033 Population Genetics | LMP1106 Molecular Techniques | LMP1107 Bioinformatics | EEB1460 Molecular Evolution | LMP1210 Biomedical Machine Learning | CHL5701 Seminars in Global Health | CHL5704 International Global Health: The Right to Health in Theory and Practice

Master's of Science Degree (M.Sc.) Biomedical Informatics — 2019 - 2020
Department of Biomedical & Molecular Sciences | School of Computing | Queen's University | Kingston, ON

- **Cumulative GPA**: 3.67/4.0
- **Supervisor**: Amber Simpson
- **Master's Thesis**: Investigating The Role of Immune Microenvironments and Quantitative Imaging Phenotypes in Pancreatic Cancer
- **Relevant Coursework**: BMIF801Processing Biomedical Data | BMIF802 Biomedical Data Analysis | BMIF803 Data Mining and Applications | PATH828 Bioinformatics Cancer Research

Honours Bachelor of Science (B.Sc.) Combined Specialization in Biology and Forensic Science — 2015 - 2019
School of Natural Sciences | Laurentian University | Sudbury, ON

- **Cumulative GPA**: 3.43/4.0
- **Supervisor**: Kabwe Nkongolo
- **Thesis**: Development and Characterization of Species Diagnostic ISSR and SCAR DNA Markers for Differentiating Red Maple (*Acer Rubrum*) and Silver Maple (*A. Saccharinum*)

- **Relevant Coursework**: CHMI2227 Biochemistry | BIOL1506 & BIOL1507 Biology I & II | BIOL2126 Cell Biology | BIOL2026 Microbiology | BIOL2007 Genetics | BIOL4801 Advanced Genetics| fors4206 Population genetics | PSYC1105 Psychology | BIOL3327 Experimental Methods in Biology | STAT2246 Statistics for Scientists

Research Experience

Doctoral Researcher 2021 - Present

Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health | Toronto ON

- Identified novel associations between exome-wide tandem repeats (TRs) and subcortical brain volumes, investigating the role of understudied genetic structural variation in brain morphology and highlighting TRs with potential functional effects on protein folding.
- Characterized genetic relationships and investigated potential causal mechanisms using regression analysis, Mendelian Randomization (MR), Genomic Structural Equation Modeling (GSEM), linkage disequilibrium score regression (LDSC), and post-GWAS analyses to better understand the genetic architecture of complex traits and disease.
- Integrated large-scale genetic, health, lifestyle, and imaging-derived phenotypes from UK Biobank and All of Us to investigate the contributions of genetic variation and modifiable risk factors to brain health and disease.
- Implemented quality control and downstream analysis of next-generation sequencing (NGS) data, processing short-read sequencing files, including BAM and VCF formats, in Linux and Bash environments to support reliable genomic analyses.
- Developed reproducible computational pipelines in R and Python using genomic analysis tools to investigate genetic associations and facilitate reproducible research.

Graduate Researcher 2019-2020

Simpson Lab, School of Computing | Queen's University | Kingston ON

- Identified 48 significant CT-derived radiomic features associated with intratumoral immune-cell infiltration in 64 patients with pancreatic ductal adenocarcinoma (PDAC), demonstrating the potential of routine medical imaging to characterize tumour immune microenvironments non-invasively.
- Developed multivariable regression models linking CT radiomic features to tumour immune-cell populations, demonstrating strong associations with mature dendritic cells and cytolytic CD8+ T cells ($R^2 > 0.8$) and supporting the potential of radiomics as a cost-efficient approach to characterizing tumour biology and informing personalized treatment strategies.
- Evaluated the prognostic value of tumour immune infiltration using Cox regression, identifying associations between regulatory T cells, cytotoxic CD8+ T cells, activated cytolytic T cells, and overall survival (C-index > 0.6), highlighting the potential of immune profiling to inform patient prognosis and treatment decisions.

Undergraduate Researcher 2018 - 2019

Nkongolo Lab, School of Natural Sciences | Laurentian University | Sudbury ON

- Developed and characterized six species-diagnostic ISSR markers using PCR and gel electrophoresis across 20 populations of red maple (*Acer rubrum*) and silver maple (*A. saccharinum*), addressing the lack of validated molecular markers for distinguishing these closely related species.
- Validated the hybrid status of *Acer freemanii* from by screening diagnostic ISSR markers, confirming its genetic origins from *A. rubrum* and *A. saccharinum* and demonstrating the utility of molecular markers for hybrid identification.
- Converted species-diagnostic ISSR markers into sequence-characterized amplified region (SCAR) markers through DNA cloning, Sanger sequencing, and primer design, establishing additional

molecular tools for species certification and tracking genetic introgression in hybrid trees and populations.

Teaching & Mentorship Experience

Sessional Course Instructor

Trent University, Master of Forensic Science Program Fall 2022

FRSC5003 – Research Design and Statistics in Forensic Science

- Independently taught 24 graduate students through weekly two-hour lectures and two-hour tutorials, delivering instruction in research design and statistical methods relevant to forensic science.
- Applied student-centred, UDL-informed teaching approaches by combining lectures, RStudio demonstrations, and guided tutorials to help students engage with statistical concepts through explanation, demonstration, and practical application
- Designed course materials, assignments, and grading rubrics aligned with learning objectives, establishing transparent expectations and supporting students in developing research design and statistical analysis skills.

Guest Lecturer

Laurentian University, Master of Forensic Science Program Summer 2024

FORS5056EL – Advanced Topics in Forensic DNA Analysis

Course Instructor: Krista Currie

- Delivered a guest lecture on advanced forensic DNA analysis to graduate students, supplementing course instruction with specialized subject-matter expertise. Date presented: June 12, 2024.

Teaching Assistantships

Practical Teaching Assistant | University of Toronto, Department of Anthropology Summer 2026 & Fall 2026

ANT202 – Biological Anthropology: Human Variation and Adaptation

Course Instructor: Esteban Parra

- Led two weekly one-hour practical sessions for approximately 30 students on human evolution, genetics, and biological variation, connecting lecture concepts with guided application.
- Taught fundamental genetics concepts, including Punnett squares, population genetics, and pedigree and genealogy analysis, helping students apply principles of inheritance and genetic variation.

Practical Teaching Assistant | University of Toronto, Department of Biology Fall 2023

BIO259 – Introduction to Biological Data

Course Instructor: Alex Nguyen Ba

- Facilitated weekly two-hour practical sessions for approximately 30 students, translating biological data concepts into guided computational exercises.
- Guided students through R programming and biological data analysis, helping them apply computational methods to biological datasets and interpret analytical results.
- Administered and invigilated term tests for approximately 400 students, supporting consistent assessment delivery and examination integrity.

Tutorial & Marking Teaching Assistant | University of Toronto, Forensic Science Program Winter 2023 & Fall 2023

FSC416 – Population Genetics

Course Instructor: Frank Wendt

- Facilitated weekly office hours focused on genetics and population genetics, helping students identify and address gaps in their conceptual understanding.
- Guided students through R programming and data analysis during tutorials, reinforcing the application of computational methods to population-genetic questions.
- Evaluated assignments and provided constructive feedback on statistical analyses and interpretation, supporting students in strengthening their analytical reasoning.

Marking Teaching Assistant | University of Toronto, Forensic Science Program Winter 2023

FSC350 – Forensic Statistics

Course Instructor: Shelby Scott

- Assessed assignments for approximately 40 students, providing targeted feedback on statistical methods and interpretation to support accurate analysis and clear communication of results.

Marking Teaching Assistant | University of Toronto, Forensic Science Program Summer 2022 & Fall 2024

FSC100 – The Real CSI

Course Instructor: Shelby Scott

- Evaluated assignments and essays for approximately 100 students, providing timely, course-aligned feedback grounded in weekly lectures and readings.
- Contributed to assessment development by drafting quiz, midterm, and assignment questions designed to evaluate student understanding across course topics.
- Administered and invigilated midterm and final examinations, supporting orderly and consistent assessment procedures.

Marking Teaching Assistant | University of Toronto, Forensic Science Program Fall/Winnter 2022

FSC482 – Professional Practice in Forensic Science

Course Instructors: Tracy Rogers & Karen Woodall

- Coordinated weekly course updates and assessment activities, helping maintain clear communication of course requirements and learning expectations.
- Assessed assignments for approximately 80 students and delivered feedback to support their development of professional knowledge and skills relevant to forensic science.

Practical & Marking Teaching Assistant | University of Toronto, Forensic Science Program Fall 2022

FSC316 – Forensic Anatomy

Course Instructor: Lilianna Watamaniuk

- Supported practical demonstrations and lectures in forensic anatomy, helping students connect anatomical concepts with applied forensic contexts.
- Evaluated weekly laboratory assignments for approximately 40 students, reinforcing their understanding of anatomical concepts and laboratory material.
- Administered and invigilated midterm and final examinations, contributing to the consistent delivery of course assessments.

Head Teaching Assistant | University of Toronto, Forensic Science Program Fall/Winter 2021 - 2024

FSC239 – Introduction to Forensic Science

Course Instructor: Vivienne Luk, Frank Wendt, Rasmus Larson

- Designed assessment materials and test questions across course topics to evaluate student learning and engage students with introductory forensic science concepts.
- Coordinated communication between students and course instructors through email, Microsoft Teams, and Zoom, relaying student concerns and clarifying course expectations.
- Facilitated virtual office hours to address academic questions and reinforce students' understanding of course material.

Laboratory Teaching Assistant | University of Toronto, Forensic Science Program Fall 2021

FSC15 – Forensic Biology

Course Instructor: Nicole Novroski

- Facilitate experiential learning through interactive laboratory sessions for eight students, connecting theoretical concepts with practical forensic biology techniques.
- Demonstrated forensic serology and laboratory techniques, including DNA extraction, amplification, quantification, and interpretation, to students. Connecting molecular methods with practical forensic applications, improving students' understanding of concepts.

Mentorship

Mentor for Undergraduate Student 2023 - 2024

Forensic Science program | Mississauga, ON

- Mentored an undergraduate capstone student in FSC485, guiding the implementation of a data-processing pipeline and application of random forest and linear regression models to analyze forensic DNA profiles.
- Designed research questions and analytical methods aligned with course learning objectives, connecting computational approaches to practical applications in forensic DNA analysis.
- Coached the student in interpreting statistical and machine-learning outputs and communicating research findings, strengthening their ability to interpret analytical results and present scientific conclusions.

Mentor for Graduate Student 2022 - 2023

Department of Anthropology | Mississauga, ON

- Guided a master's student in characterizing ancestry groups within a Canadian population, supporting the investigation of population-level genetic variation through DNA profile analysis.
- Provided technical guidance in sequence alignment and FaSTR-based allele calling to facilitate DNA profile analysis and the characterization of genetic variation.

Anthropology Graduate Student Union Peer Mentor 2022 - 2023

Anthropology Graduate Student Union | Toronto, ON

- Guided a graduate student in navigating academic and departmental expectations by sharing strategies for managing coursework, accessing university resources, and understanding departmental procedures to support their academic progress and independence.

Professional Experience

Chief Presiding Officer (CPO) 2022 – Present

Office of the Registrar at University of Toronto, Mississauga | Mississauga ON

- Prepared examination rooms and organized seating arrangements and exam materials to facilitate orderly, efficient examination administration.
- Verified student identification and monitored examination rooms to uphold academic integrity, prevent academic offences, and enforce examination policies.
- Managed examination disruptions, including student illness, late arrivals, and technical or physical incidents, documenting violations and applying established protocols to maintain a fair and secure testing environment.

Clinical Trial Research Assistant 2021 - 2022

Population Health Research Institute | Hamilton ON

- Coordinated clinical trial operations across three Ontario hospital sites, liaising with internal and external stakeholders to manage study timelines, training, protocols, and budgets and support consistent trial execution.
- Identified study trends and refined 20 electronic case report forms (eCRFs) for more than 60 participants in the BRAVE study, supporting systematic clinical data collection and standardized documentation.
- Managed clinical data entry and quality control in REDCap, monitoring data accuracy and adherence to study requirements to support reliable clinical trial data and efficient study operations.

Operations Coordinator 2020 - 2021

Trillium Health Partners | Mississauga ON

- Analyzed operational data and prepared ad hoc reports to inform data-driven decisions, including staff unit transfers and resource allocation.
- Coordinated more than 200 support staff during the COVID-19 operational restructuring, contributing to stabilization of the revised support model and validation of operational workflow.
- Validated schedules for staff based on priority principles given by the senior leadership team to ensure support was delivered effectively.

Community Engagement & Professional Services

UTM Graduate Student Representative June 2026 - Present

Sexual Violence Prevention Student Advisory Committee | Mississauga, ON

- Contribute student-informed feedback during quarterly advisory committee meetings to help refine Sexual Violence Prevention Centre initiatives and strengthen support for the University of Toronto community

STEM Outreach Coordinator (Volunteer) 2022 - Present

Canadian Girls in Science (CAGIS) | Mississauga, ON

- Organized and facilitated monthly events tailored to engage girls, nonbinary, and gender non-conforming youth aged 7-17, fostering interest and participation in science, technology, engineering, and mathematics fields.
- Developed and implemented a workshop focused on creating machine learning models using Teachable Machine and Scratch platforms, enabling participants to learn about and apply machine learning concepts in a hands-on, interactive environment.

Social Coordinator 2022 - 2023

Anthropology Graduate Student Union | Toronto, ON

- Executed bi-monthly social events for the Anthropology Graduate Student Union, such as trivia nights, fostering a sense of community and camaraderie among graduate students in the department.
- Coordinated logistics and venue arrangements for each event, ensuring smooth execution and adherence to budgetary constraints while maximizing attendance and engagement.

Peer Review Publications

Published

1. **Panoyan MA**., & Wendt F. (2023). The role of tandem repeat expansions in brain disorders. Emerging Topics in Life Sciences. doi: 10.1042/ETLS20230022 (**Impact Factor: 3.3**)
2. Shadoff R, **Panoyan MA**., & Novroski N. (2022). Microbial Forensics: A present to future perspective on genomic targets, bioinformatic challenges, and applications. Forensic Genomics, 2(2), 42–64. doi: 10.1089/forensic.2022.0007 (**Impact Factor: 0.37**)

3. **Panoyan MA***, Boyd M*, Michael P, & Nkongolo KK. 2019. Development and characterization of species diagnostic ISSR and SCAR DNA markers for differentiating red maple (*Acer rubrum*) and silver maple (*A. saccharinum*). Genome 62(8):527-535. doi: 10.1139/gen-2019-0037 * co-first authors (**Impact Factor: 2**)

Under Review

1. **Panoyan MA**., Shi Y., Abbatangelo C.L., Adler N., Moo-Choy A., Parra E., ... & Felsky D. (2023). Exome-wide tandem repeats confer large effects on subcortical volumes in UK Biobank participants. medRxiv. https://doi.org/10.1101/2023.12.11.23299818 (**Molecular Psychiatry, Impact Factor: 10.4**)

In Preparation

2. **Panoyan MA**., Wendt FR., Parra EJ., & Felsky D. (2026). Investigating the relationship between subcortical brain volume and alcohol consumption using exome-wide tandem repeats.
3. **Panoyan MA**., Shi Y., Polimanti R., & Wendt FR. (2026). Detecting tandem repeats associated with alcohol use and abuse.

Conferences

Oral Presentations

1. **Panoyan MA**, et al. (2026). *Investigating the causal relationship between cardiometabolic traits and Alzheimer's disease.* Novo Nordisk Healthy Population Discovery Summit, Mississauga, ON, Canada. June 2026.
2. **Panoyan MA**, et al. (2023). *Investigating the role of exome-wide tandem repeats in alcohol use and misuse.* World Congress of Psychiatric Genetics (WCPG), Montreal, QC, Canada. October 2023.
3. **Panoyan MA, Boyd M**, Michael P, & Nkongolo KK. (2019). *Development and characterization of species-diagnostic ISSR and SCAR DNA markers for differentiating red maple (Acer rubrum) and silver maple (A. saccharinum).* Ontario Biology Day Student Conference, London, ON, Canada. March 2019

Poster Presentations

1. **Panoyan MA,** et al. (2024). *Investigating the relationship between subcortical brain volume and alcohol consumption using exome-wide tandem repeats.* World Congress of Psychiatric Genetics (WCPG), Singapore. October 2024.
2. **Panoyan MA,** et al. (2020). *Investigating the role of immune environments and quantitative imaging phenotypes in pancreatic cancer.* International Molecular and Nanomedicine Organization (IMNO) Conference, Virtual. March 2020.
3. **Panoyan MA,** et al. (2019). *Investigating the role of immune environments and quantitative imaging phenotypes in pancreatic cancer.* CSearch Conference, Queen's University, Kingston, ON, Canada. October 2019.

Competitions

1. **Panoyan MA** & Novroski N. (2022). *Classification of ancestry by supervised machine learning models on forensically relevant SNPs.* ISHI Three-Minute Thesis Competition, Virtual. June 2022. Video presentation.

Grants

Doctoral Completion Award |$9,100	2026
Ontario graduate scholarship | $15,000	2024
Novo Nordisk Network for Healthy Populations Graduate Award | $20,000	2024
SGS Conference Grant | $1,230	2024

Pilot Research Award | $4,000 2023
GEF Graduate Fund Award | $650 2022

Reviewer Activities

Reviewer | Translational Psychiatry Journal Winter 2024

Professional Memberships

Women's Health Research Cluster Member 2024 - Present
Novo Nordisk Network for Healthy Populations Member 2024 - Present
International Society of Psychiatric Genetics (ISPG) Student Member 2023 - Present

Certifications

- Course Design | Teaching Assistant Training Program (TATP), University of Toronto In Progress
- Educational Technology | Teaching Assistant Training Program (TATP), University of Toronto In Progress
- Accessible and Inclusive Teaching | Teaching Assistant Training Program (TATP), University of Toronto In Progress
- Foundations of Teaching in Higher Education | Teaching Assistant Training Program (TATP), University of Toronto In Progress
- Building a Culture of Consent at the University of Toronto: Consent, Communication, and Sexual Violence Prevention Education Module for Students | Sexual Violence Prevention & Support Centre (SVPSC), University of Toronto 2026
- CITI Canada Privacy Course 2021
- Good Clinical Practice (GCP) 2021
- Health Canada Division 5 Drugs for Clinical Trials Involving Humans 2021
- Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans Course on Research Ethics (TCPS 2: CORE) 2021
- Forensic Identification Certificate 2019